

2MVA Level I Multivariable & Vector Analysis (25667)

ASSESSED PROBLEM SHEET 3

The submission deadline is 5:00pm on Wednesday 3 December 2025
(Answer all Questions)

Question 1:

(a) What is the length of the curve C which is the segment of a circular helix parameterized by

$$C = (x(t), y(t), z(t)) = (a \cos(t), a \sin(t), bt) \quad \text{for } 0 \leq t \leq 3\pi$$

where a and b are positive constants.

(b) What is the mass of this helical segment if its density (mass per unit length) is given by $\rho(x, y, z) = x^2 + y^2 + z^2$?

Question 2: Let $\mathbf{F} = (y^3 + y, 3xy^2)$ and let $\mathbf{r} = (x, y)$ be the position vector in $\mathbb{R}^2$.

(a) Is $\mathbf{F}$ a conservative field? Explain your answer.

(b) Let $C = C_1 + C_2$ be closed curve where C_1 is the semicircle along $x^2 + y^2 = 1$ traversed in the counter-clockwise direction from the point $(1, 0)$ to the point $(-1, 0)$ and C_2 is the straight line segment from $(-1, 0)$ to $(1, 0)$. Using Green's Theorem or otherwise calculate $\oint_C \mathbf{F} \cdot d\mathbf{r}$.

Question 3: Let $\mathbf{F} = (x + y + z, \sin(xz), yx)$ in $\mathbb{R}^3$. Calculate $\text{curl}(\mathbf{F}) = \nabla \times \mathbf{F}$.

Question 4: The field $\mathbf{F} = (2x + \frac{1}{y}, 2y - \frac{x}{y^2})$ is a gradient field in $\mathbb{R}^2$. Determine the scalar function $f = f(x, y)$ such that $\mathbf{F} = \nabla f$. Calculate $\int_C \mathbf{F} \cdot d\mathbf{r}$ where C is the straight line segment from $(1, 1)$ to $(2, 2)$.